Assignment 2

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Question 1

Q1. Using the provided random number generator (`bad_rnd.f90`), generate 1000 random numbers. Plot a scatter plot of successive pairs (x_n, x_{n+1}) to examine whether the numbers are correlated. Comment on the quality of the random number generator.

Answer. Using the program `bad_rnd.f90`, a sequence of 5000 random numbers $\{x_1, x_2, \ldots, x_{5000}\}$ was generated.

The sample mean and standard deviation of the data were found to be

$$\langle x \rangle = 0.5006, \qquad \sigma = 0.2867.$$

These values are close to the theoretical mean and standard deviation of a uniform distribution on $[0, 1]$, for which

$$\langle x \rangle_{\text{theory}} = \frac{1}{2}, \qquad \sigma_{\text{theory}} = \frac{1}{\sqrt{12}} \approx 0.2887.$$

This indicates that the generator reproduces the correct one-point statistics.
To test for correlations, a scatter plot of successive values (x_n, x_{n+1}) was constructed.

If the random number generator were ideal, the points in this scatter plot would be uniformly distributed over the unit square, showing no visible structure or pattern. However, the scatter plot obtained from `bad_rnd.f90` shows clear correlations, with points lying along distinct lines or exhibiting clustering rather than filling the plane uniformly.

This behavior indicates that successive random numbers are not statistically independent. Such correlations are characteristic of a poor-quality random number generator.

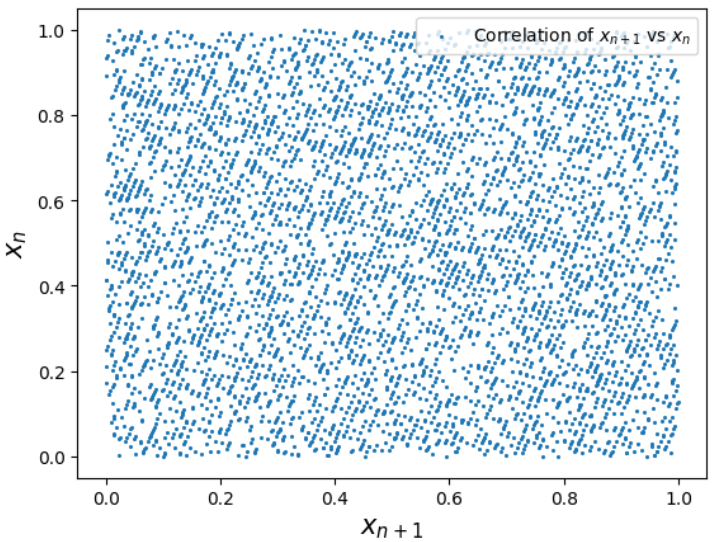

Figure 1: Correlation for bad RNG

Additionally, the autocorrelation function was computed as

$$C(k) = \frac{\langle x_n x_{n+k} \rangle - \langle x \rangle^2}{\sigma^2},$$

where k is the lag and σ is the standard deviation of the data. The autocorrelation plot shows non-zero values for small lags, whereas a high-quality random number generator would produce correlations consistent with zero for all non-zero lags.

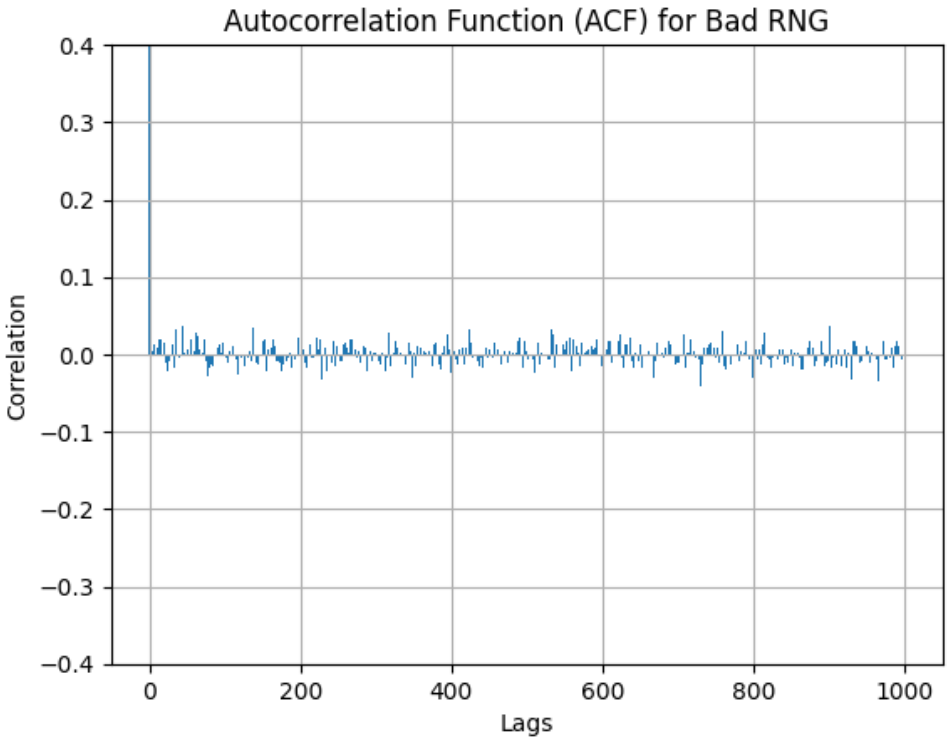

Figure 2: Autocorrelation for bad RNG

Question 2

Use a random number generator to generate uniform random numbers in the interval $[0, 1]$. Verify the quality of the generator by analyzing the probability distribution, correlations, and statistical moments of the generated data.

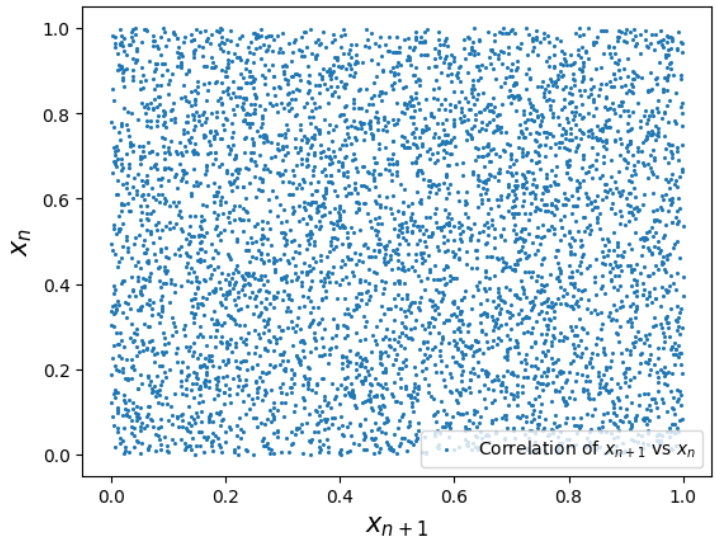

Figure 3: Correlation scatterplot for good RNG

Answer. A sequence of N random numbers $\{x_1, x_2, \ldots, x_N\}$ uniformly distributed in the interval $[0, 1]$ was generated using a random number generator.

- **Probability distribution:** A histogram of the generated data was plotted and found to be approximately flat over the interval $[0, 1]$, confirming that the numbers follow a uniform probability distribution.

- **Scatter plot:** A scatter plot of successive values (x_n, x_{n+1}) was constructed. The points were uniformly distributed over the unit square with no visible structure, indicating that successive random numbers are uncorrelated.

- **Correlation function:** The autocorrelation function

$$C(k) = \frac{\langle x_n x_{n+k} \rangle - \langle x \rangle^2}{\sigma^2}$$

  was calculated for different lags k. The correlation function was found to be close to zero for all non-zero lags, confirming the absence of correlations.

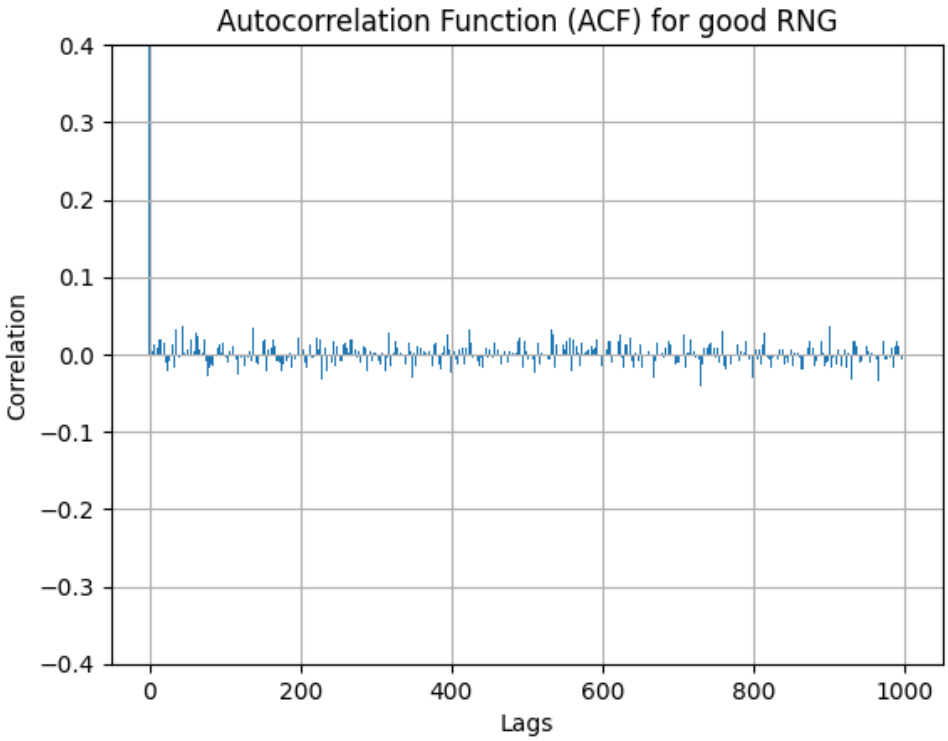

Figure 4: Autocorrelation plot for good RNG

- **Standard deviation:** The mean and standard deviation of the data were computed as

$$\langle x \rangle \approx \frac{1}{2}, \qquad \sigma \approx \frac{1}{\sqrt{12}},$$

in agreement with the theoretical values for a uniform distribution on $[0, 1]$.

Question 3

Generate random numbers with (a) an exponential distribution $p(x) = e^{-2x}$, and (b) a Gaussian distribution with standard deviation $\sigma = 2$.

Answer. Random numbers from non-uniform distributions can be generated starting from uniformly distributed random numbers using appropriate transformations.

(a) **Exponential distribution**

The target probability density is

$$p(x) = 2e^{-2x}, \qquad x \geq 0,$$

where the factor 2 ensures normalization.

The cumulative distribution function (CDF) is

$$F(x) = \int_0^x 2e^{-2x'}\, dx' = 1 - e^{-2x}.$$

Using the inverse transform method, a uniform random number $u \in (0, 1)$ is related to x by

$$u = 1 - e^{-2x}.$$

Solving for x gives

$$x = -\frac{1}{2}\ln(1 - u).$$

Since $1 - u$ is also uniformly distributed in $(0, 1)$, this is commonly written as

$$x = -\frac{1}{2}\ln u.$$

This transformation generates random numbers with the desired exponential distribution. When fitted to the curve for exponential relation we get a parameter value of -2 for the exponent as expected.

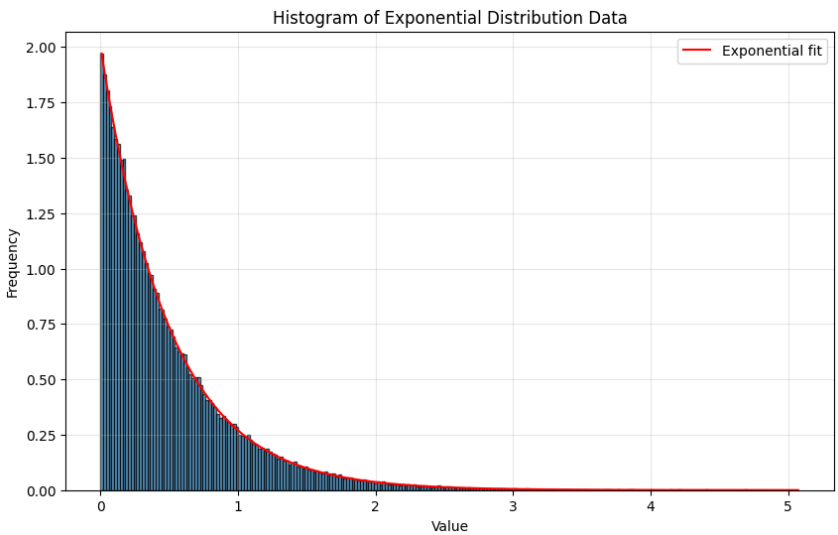

Figure 5: Exponential distribution

(b) **Gaussian distribution**

To generate Gaussian random numbers, the Box–Muller transformation is used. Two independent uniform random numbers $u_1, u_2 \in (0, 1)$ are generated and transformed as

$$z = \sqrt{-2\ln u_1}\cos(2\pi u_2),$$

which produces a standard normal random variable $z \sim \mathcal{N}(0, 1)$.

To obtain a Gaussian distribution with standard deviation $\sigma = 2$, the variable is scaled as

$$x = 2z.$$

The resulting distribution has mean zero and standard deviation $\sigma = 2$. When fitted to curve of similar nature we obtain parameter of $\sigma = 1.993$ which is very close to the expected Std. of the distribution.

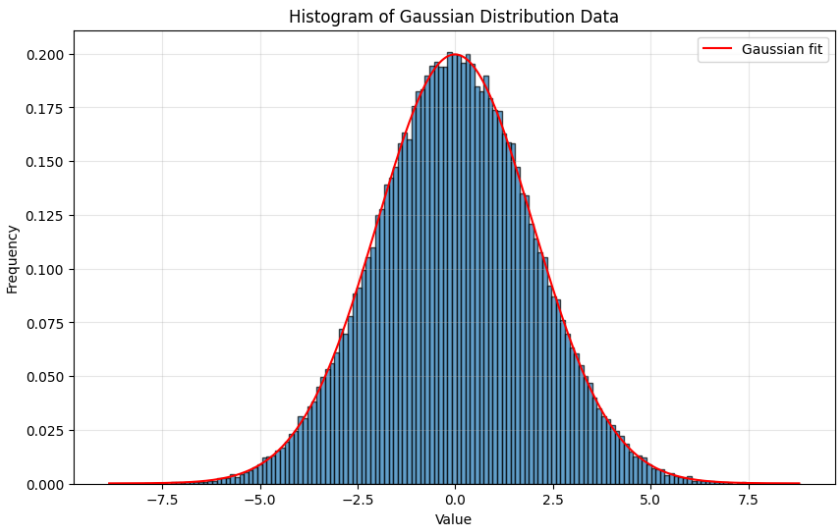

Figure 6: Gaussian distribution

Question 4

Using the trapezoidal rule, evaluate the integral

$$\int_0^1 \frac{4}{1+x^2}\,dx$$

to determine the value of π. Compare the computed value with the actual value of π, where the actual value is obtained from

$$\pi = 4\tan^{-1}(1).$$

Use $n = 10^2, 10^3, \ldots, 10^9$ grid points and perform the calculation using both single and double precision. Plot the absolute error versus n on a log–log scale. Comment on the observed behavior and determine the slope of the fitted line.

Answer. The integral

$$\int_0^1 \frac{4}{1+x^2}\,dx$$

has the exact analytical value

$$4\tan^{-1}(1) = \pi.$$

This identity allows π to be computed numerically without directly using a predefined value of π such as $22/7$, which is only an approximation and would invalidate the numerical comparison.

Trapezoidal rule

The interval $[0, 1]$ is divided into n equal subintervals of width

$$h = \frac{1}{n}.$$

The trapezoidal approximation is given by

$$I_n = h\left[\frac{1}{2}f(x_0) + \sum_{i=1}^{n-1} f(x_i) + \frac{1}{2}f(x_n)\right], \quad f(x) = \frac{4}{1+x^2}.$$

Error analysis

The absolute error in the numerical estimate of π is defined as

$$\epsilon(n) = |I_n - \pi_{\text{actual}}|, \qquad \pi_{\text{actual}} = 4\tan^{-1}(1).$$

The trapezoidal rule has a theoretical global error scaling as

$$\epsilon(n) \sim \mathcal{O}(h^2) = \mathcal{O}(n^{-2}).$$

Log–log plot

A log–log plot of $\epsilon(n)$ versus n was constructed for $n = 10^2, 10^3, \ldots, 10^9$.

- In double precision, the error decreases approximately as $\epsilon(n) \propto n^{-2}$ over a wide range of n.
- A linear fit in the scaling region yields a slope close to -2, consistent with the theoretical expectation for the trapezoidal rule.
- For very large n, the error saturates and may increase due to round-off errors, especially in single precision.
- In single precision, loss of accuracy occurs at much smaller n compared to double precision because of limited floating-point resolution.

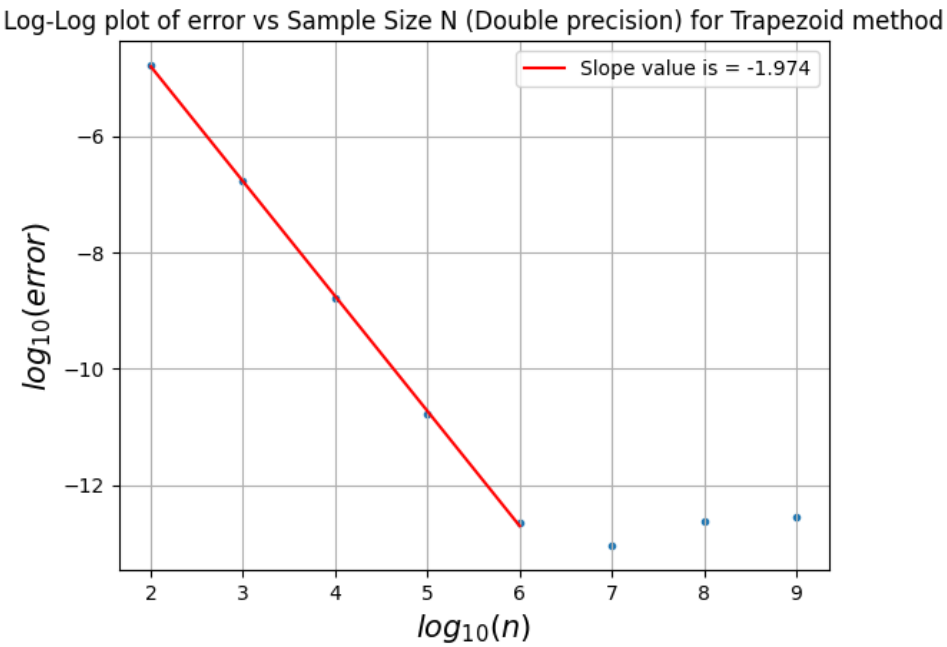

Figure 7: Double precision variable for trapezoid method

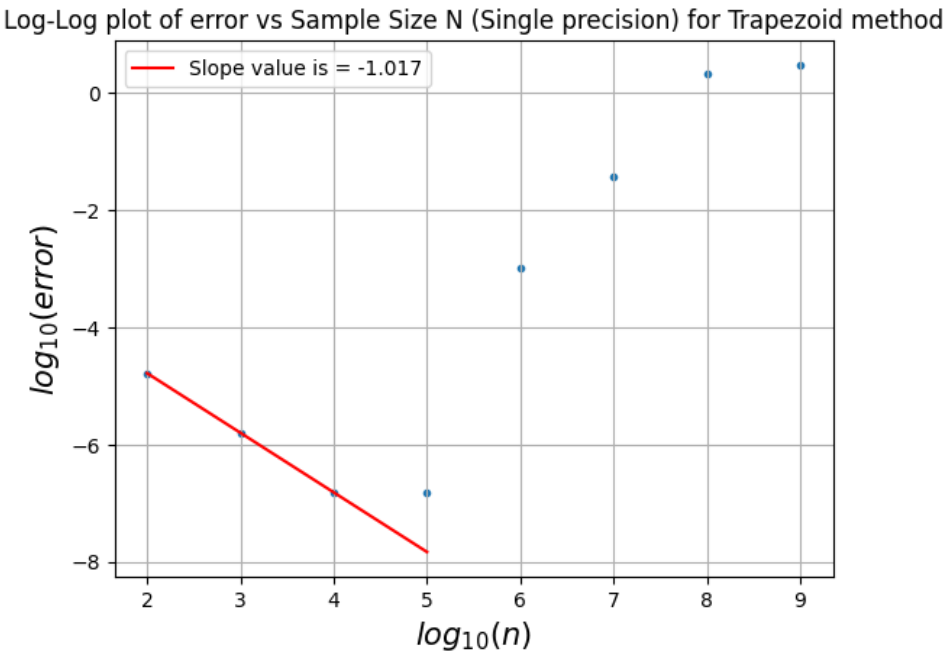

Figure 8: Single precision variable for trapezoid method

Question 5

Estimate the value of π using Buffon's needle experiment. Consider a grid of parallel lines separated by a distance d. Needles of length l ($l < d$) are dropped randomly onto the grid. The probability that a needle intersects a line is

$$P = \frac{2l}{\pi d}.$$

Using a random number generator, simulate this experiment and estimate π. Vary the number of throws n_{throw} and plot the absolute error versus n_{throw}. Comment on the error scaling and compare the efficiency of this method with the trapezoidal-rule method used in Q4.

Answer. In Buffon's needle experiment, the probability that a needle of length l intersects one of the parallel lines separated by distance d is

$$P = \frac{2l}{\pi d}.$$

Rearranging, an estimate of π is obtained as

$$\pi_{\text{est}} = \frac{2l}{d} \frac{n_{\text{throw}}}{n_{\text{hit}}},$$

where n_{hit} is the number of times the needle intersects a line during n_{throw} trials.

Simulation method

For each throw, two independent random variables are generated:

- The distance $x \in [0, d/2]$ from the center of the needle to the nearest line.
- The orientation angle $\theta \in [0, \pi/2]$ between the needle and the normal to the lines.

A needle intersects a line if

$$x \leq \frac{l}{2} \cos\theta.$$

Counting the number of hits over many throws yields an estimate of the probability $P = n_{\text{hit}}/n_{\text{throw}}$, from which π is computed.

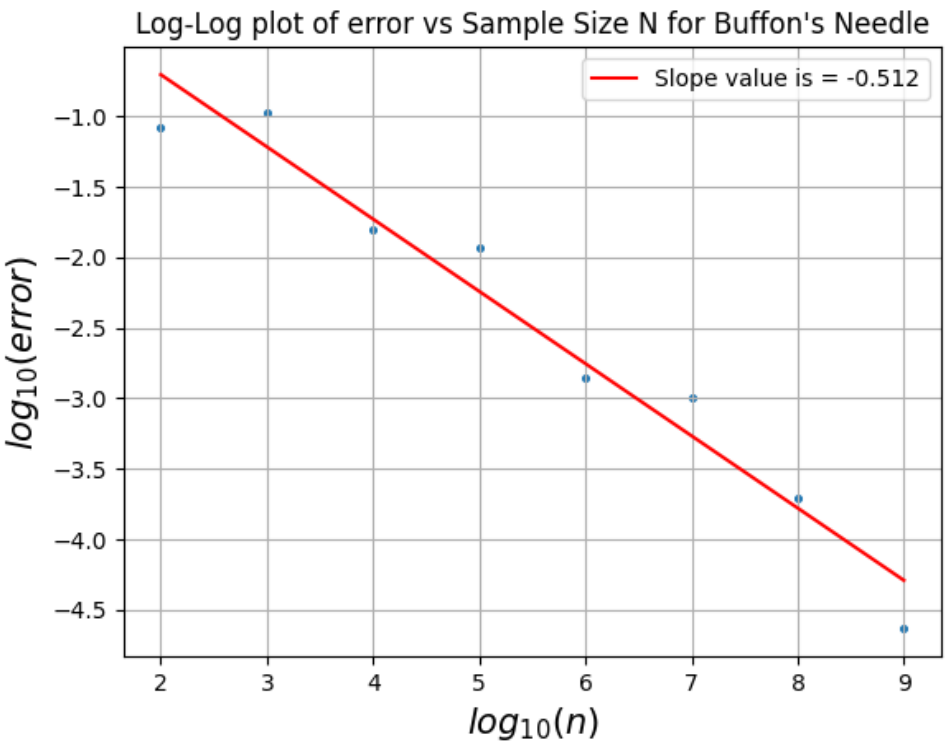

Figure 9: Log plot error vs sample size for buffon's needle

Error analysis

The absolute error is defined as

$$\epsilon(n_{\text{throw}}) = |\pi_{\text{est}} - \pi_{\text{actual}}|\,, \qquad \pi_{\text{actual}} = 4\tan^{-1}(1).$$

A log–log plot of ϵ versus n_{throw} shows that the error decreases approximately as

$$\epsilon \propto n_{\text{throw}}^{-1/2}.$$

This behavior is expected, since the Buffon's needle method is a Monte Carlo technique and statistical errors scale as $1/\sqrt{N}$.

Comparison with trapezoidal method

The trapezoidal-rule integration used in Q4 has a deterministic error scaling of

$$\epsilon \propto n^{-2},$$

which converges much faster than the Monte Carlo scaling of $n^{-1/2}$ observed in Buffon's needle experiment.

Question 6

Compute the electron–electron interaction (correlation) energy of the helium atom assuming both electrons occupy hydrogenic $1s$-like orbitals. The single-electron wavefunction is

$$\psi_{1s}(r_i) = \alpha e^{-\alpha r_i}, \qquad \alpha = 2,$$

where $r_i = \sqrt{x_i^2 + y_i^2 + z_i^2}$. The two-electron wavefunction is taken as

$$\psi(\mathbf{r}_1, \mathbf{r}_2) = Ae^{-\alpha(r_1+r_2)}.$$

where the value of A - normalization constant calculated from perturbation theory is:

$$A = \frac{\alpha^3}{\pi}$$

The expectation value of the electron–electron interaction is

$$\left\langle \frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} \right\rangle = A^2 \int d\mathbf{r}_1\, d\mathbf{r}_2\, e^{-2\alpha(r_1+r_2)}\, \frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|}.$$

This integral is evaluated using (a) brute-force Monte Carlo integration and (b) importance sampling Monte Carlo integration.

Coordinate transformation

Switching to spherical coordinates for each electron,

$$d\mathbf{r}_1 d\mathbf{r}_2 = r_1^2 r_2^2\, dr_1 dr_2\, d(\cos\theta_1) d(\cos\theta_2)\, d\phi_1 d\phi_2.$$

The inter-electron distance is

$$r_{12} = |\mathbf{r}_1 - \mathbf{r}_2| = \sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos\beta},$$

with

$$\cos\beta = \cos\theta_1 \cos\theta_2 + \sin\theta_1 \sin\theta_2 \cos(\phi_1 - \phi_2).$$

We use a sample size of 10^6 for both simulations:

(a) Brute-force Monte Carlo

In the brute-force approach, all six variables $(r_1, r_2, \cos\theta_1, \cos\theta_2, \phi_1, \phi_2)$ are sampled uniformly over their respective domains. The Monte Carlo estimator is

$$I \approx \frac{V}{N} A^2 \sum_{i=1}^{N} \frac{r_1^2 r_2^2}{r_{12}} e^{-2\alpha(r_1+r_2)},$$

where V is the total integration volume and N is 10000000
The statistical error is estimated from the variance of the integrand:

$$\sigma = \sqrt{\frac{\langle f^2 \rangle - \langle f \rangle^2}{N}},$$

which is valid even when the exact value of the integral is unknown.
Brute-force result:

$$\langle r_{12}^{-1} \rangle = 1.2497 \pm 0.0021.$$

(b) Importance sampling Monte Carlo

The dominant radial dependence of the integrand is

$$r^2 e^{-2\alpha r}.$$

Therefore, an efficient choice for importance sampling is the radial probability density

$$p(r) = 4\alpha^3 r^2 e^{-2\alpha r},$$

which is a Gamma distribution with shape parameter $k = 3$ and rate 2α.
A Gamma-distributed random variable with this distribution can be generated by summing three independent exponential variables:

$$r = \frac{-\ln u_1 - \ln u_2 - \ln u_3}{2\alpha},$$

where $u_1, u_2, u_3 \in (0, 1)$ are uniformly distributed random numbers.
Angular variables are sampled uniformly as in the brute-force method. With this choice of sampling distribution, both the exponential factors and the Jacobian cancel exactly, and the estimator simplifies to

$$I \approx \left\langle \frac{1}{r_{12}} \right\rangle.$$

Importance sampling result

Using the corrected Gamma-distribution sampling for the radial coordinates, the Monte Carlo simulation yields for $N = 10000000$

$$\boxed{\langle r_{12}^{-1} \rangle = 1.2501 \pm 0.00033}$$

where the uncertainty is the standard Monte Carlo error $\sigma \propto 1/\sqrt{N}$.

Error estimation without exact value

When the exact value of the integral is not known, the error is estimated from the variance of the Monte Carlo estimator:

$$\sigma = \sqrt{\frac{\mathrm{Var}(f)}{N}}.$$

This provides a reliable statistical uncertainty under the assumption of independent samples.

Efficiency comparison

Both brute-force and importance sampling Monte Carlo methods exhibit the same asymptotic error scaling,

$$\sigma \sim N^{-1/2}.$$

However, importance sampling dramatically reduces the variance by concentrating sampling in regions where the integrand is largest. Here we can clearly see that σ is an order of magnitude lower for importance sampling than brute force method for the same N.
Conclusion: The brute-force method wastes most samples in regions where the integrand is exponentially small, leading to slow convergence. Importance sampling using the correct Gamma-distributed radial sampling is significantly more efficient and produces a much smaller error for the same number of Monte Carlo samples.